25901157 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSECHEA03 - MANUFACTURE OF PETROCHEMICALS I

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **4** marks

1. State major engineering problems in Freush process. [K] / [CO1]
2. Discuss the role of monoethanol amine (MEA) in hydrogen purification. [U] / [CO1]
3. What are branched polymers? Give examples. [K] / [CO2]
4. List the uses of plastics. [K] / [CO3]
5. Discuss on reinforcements used in plastics and their advantages. [U] / [CO3]
6. List any four properties of elastomers. [K] / [CO4]
7. Describe nitrile rubber. [U] / [CO4]

[4x5 = 20]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **7** marks

8. Outline emulsion polymerization with its components. [An] / [CO4]

9. Explain the manufacture of nitro benzene with reaction. [U] / [CO1]

10. Explain the properties and applications of thermoplastics and thermosetting plastics. [U] / [CO3]

11. Explain the preparation of nylon-6,6 using hexamethylene diamine and adipic acid with a neat reaction. [U] / [CO2]

12. Describe the advantages and applications of transfer moulding. [U] / [CO3]

13. Using the properties of PAN, explain its suitability for fibers, filtration membranes, and protective materials. [A] / [CO2]

14. Describe preparation, properties, and uses of nitroethane, 1-nitropropane and 2-nitropropane. [U] / [CO1]

[7x5 = 35]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

15. Describe the manufacture of polystyrene by bulk polymerisation. Explain the preparation steps, properties, and applications of polystyrene. [A] / [CO2]

16. Illustrate fibres and explain fibre spinning techniques. [A] / [CO4]

[15x1 = 15]